

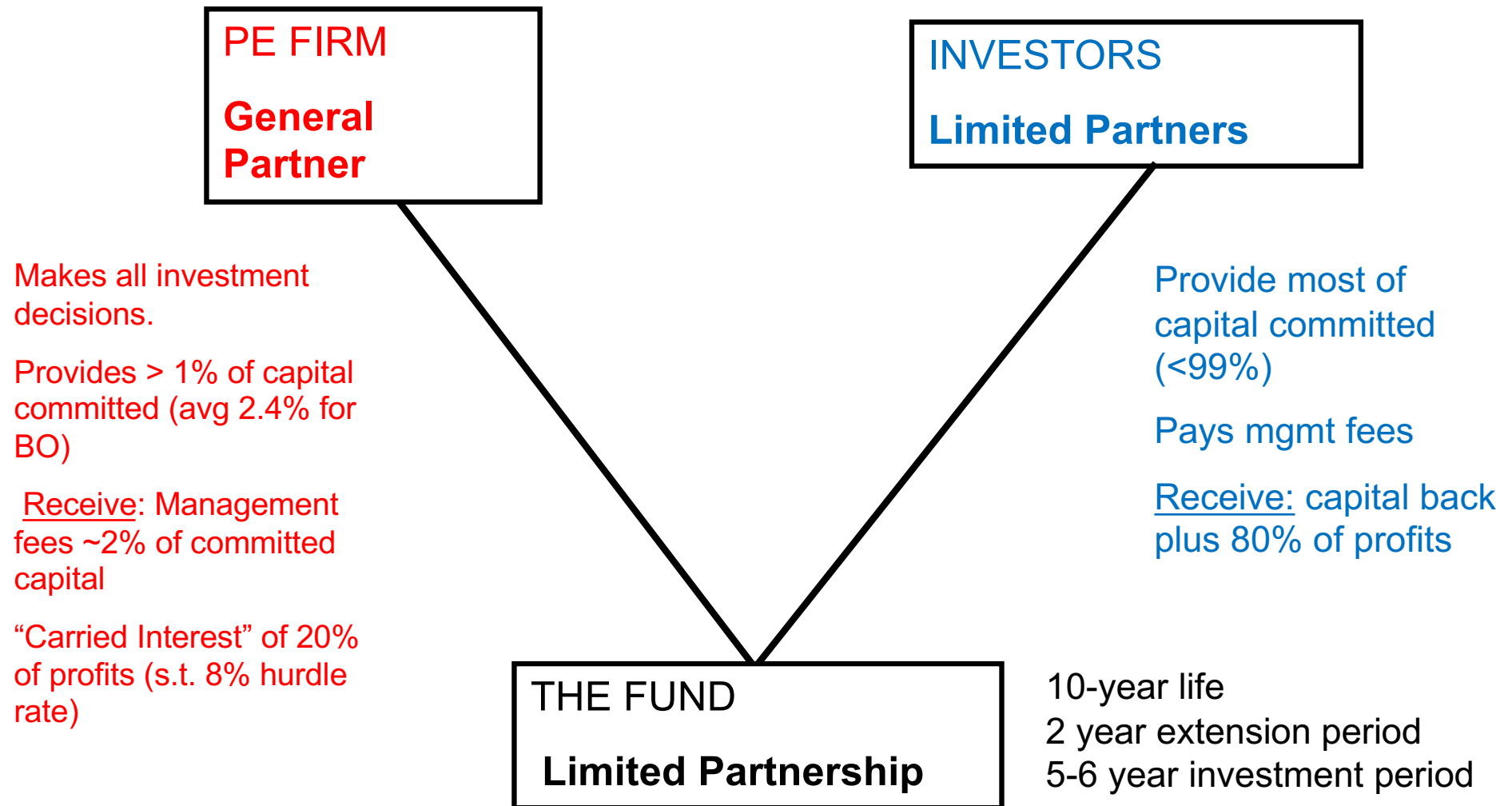


Thule *Class Slides*

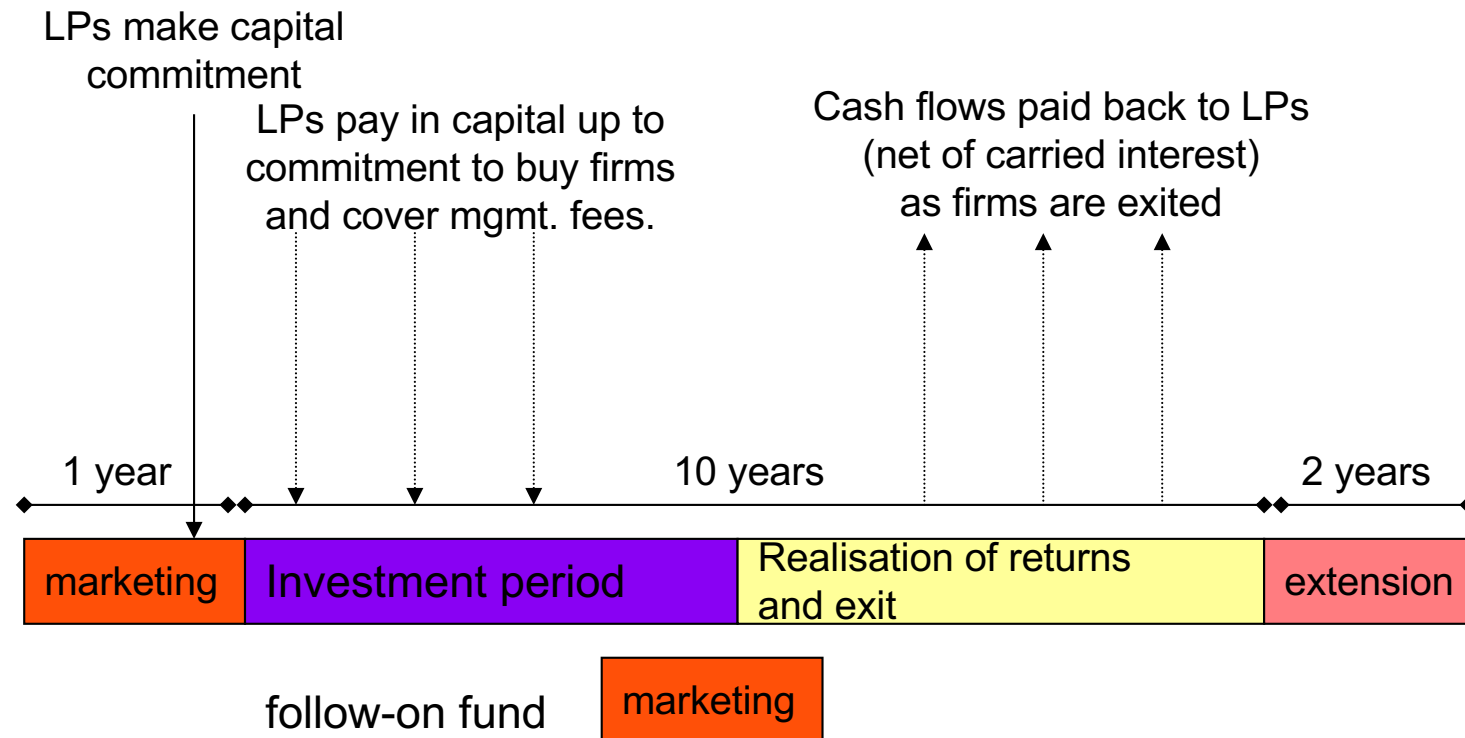
Financial Strategy
Winter 2026

Per Strömberg
University of Chicago
Booth School of Business

Structure of PE funds (I)

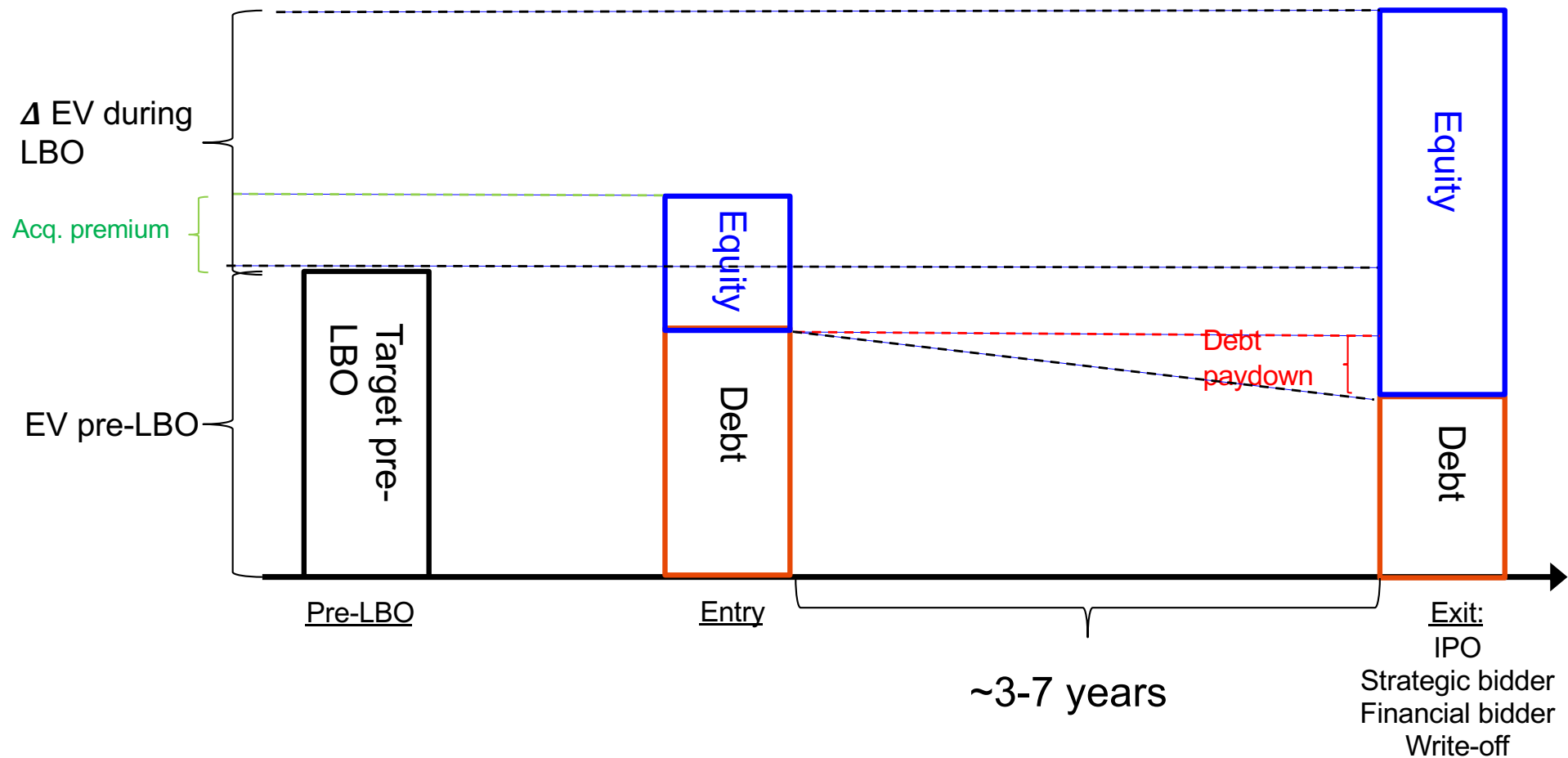


Structure of PE funds (II)

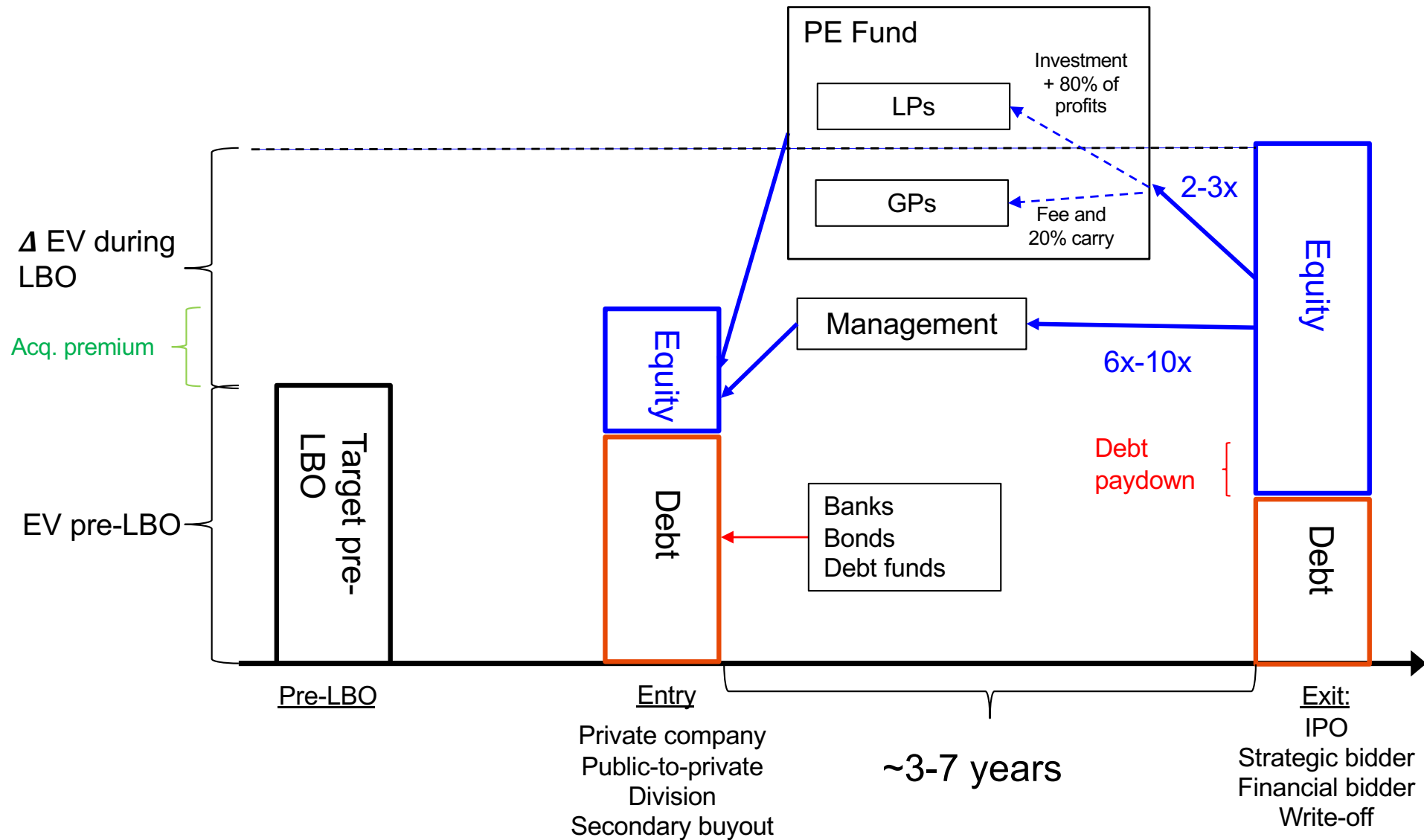


- Given lack of liquidity, LPs expect net returns 3-4% above public equity *net of fees*
- Management fee + carried interest → total fees of 5-7% per year

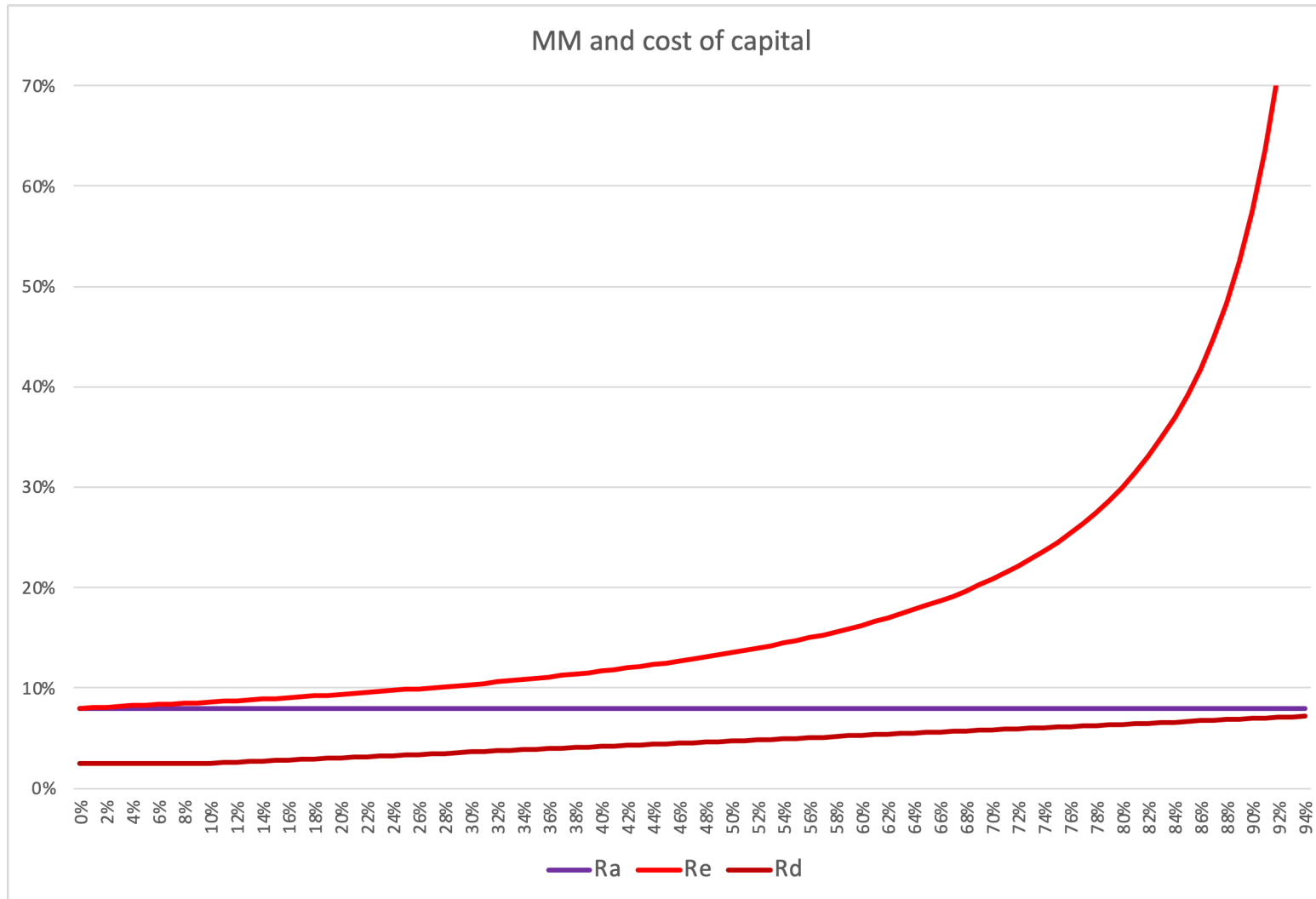
The PE investment model



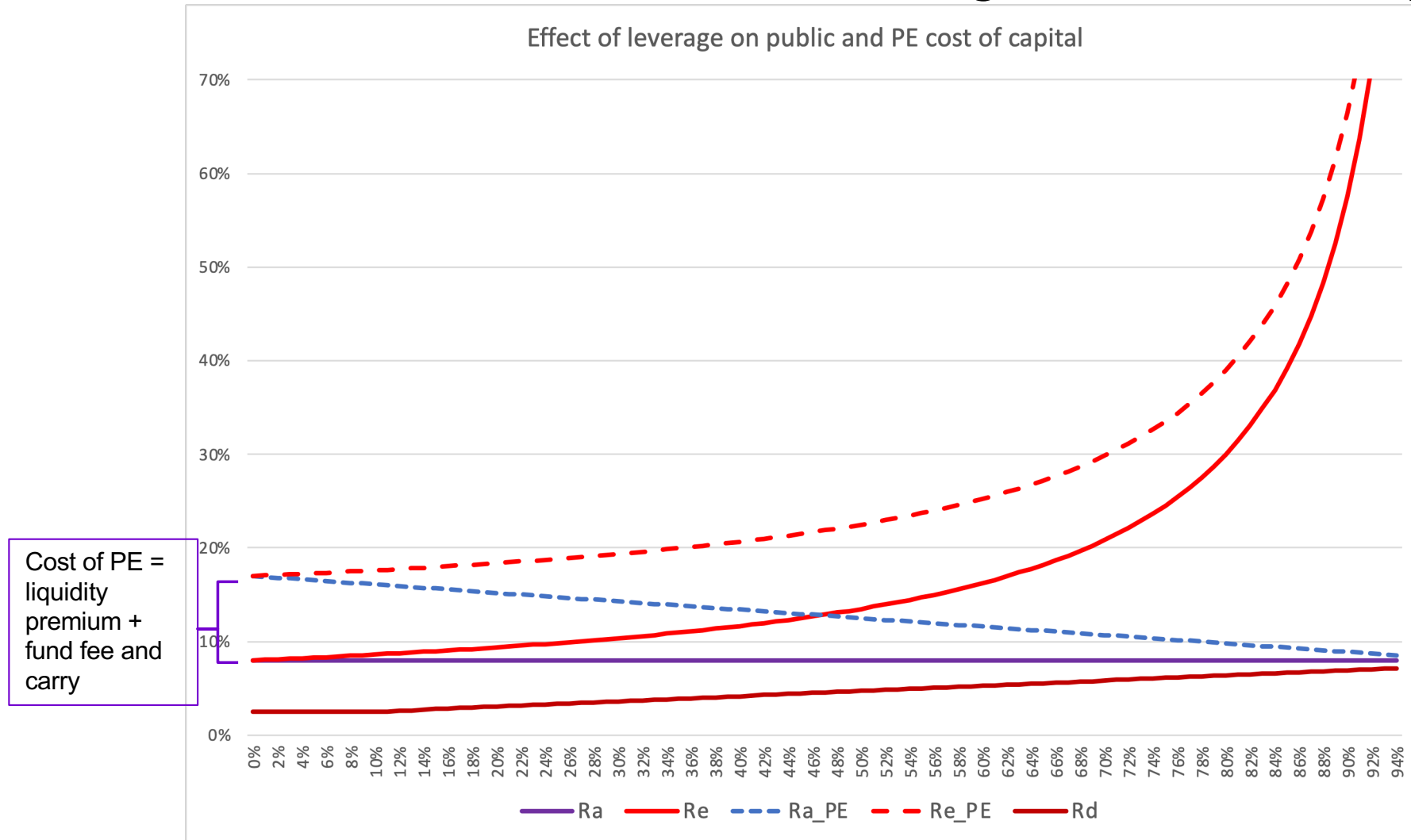
The PE investment model



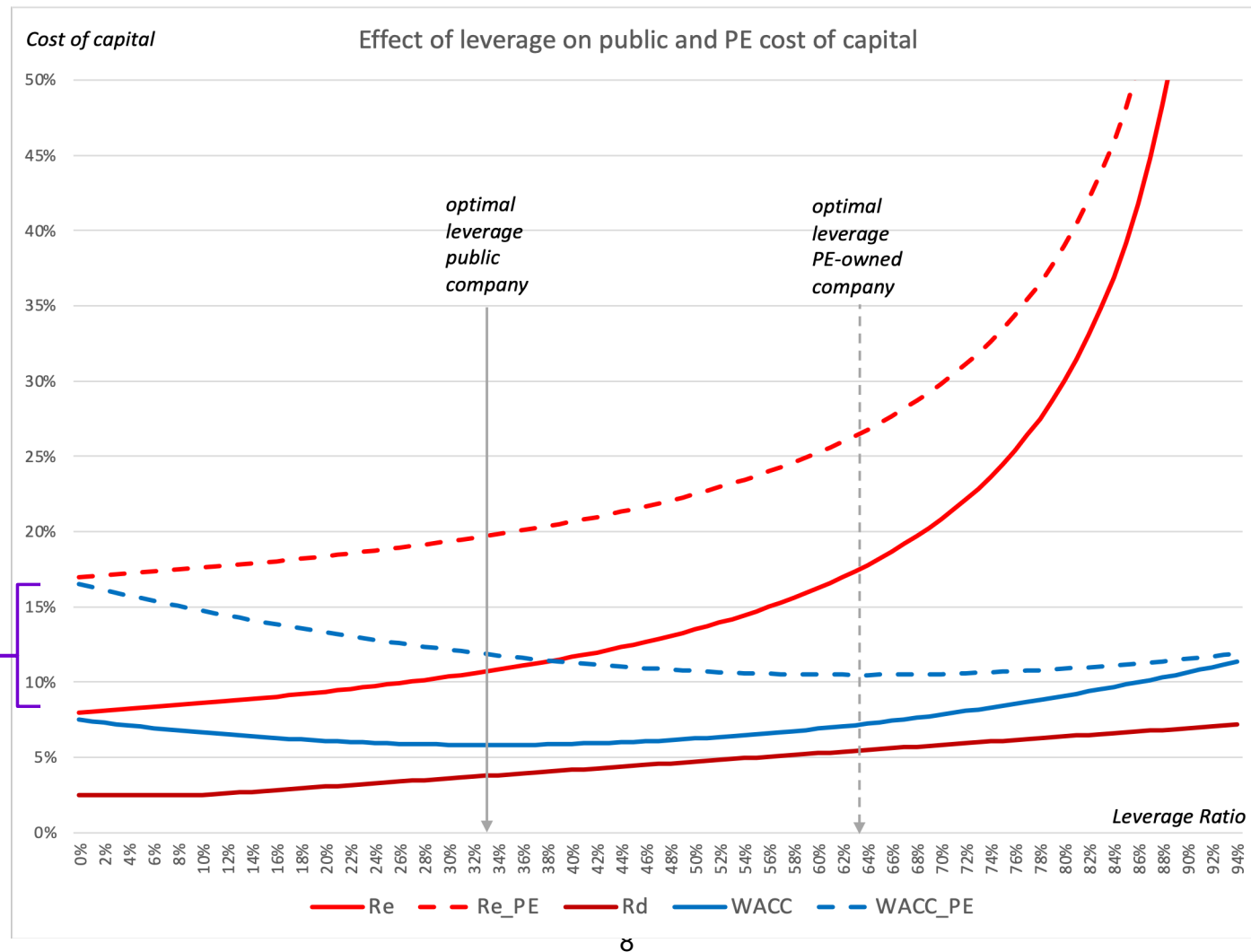
For listed companies, weighted average cost of capital does not change with leverage (MM Proposition II)...



...but for PE it does!
 In an MM world, more leverage decreases R_A



With costs and benefits of debt ($PVTS + PV(Inc) - PV(COFD)$), optimal leverage still higher in PE





Thule

Why This Case?

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Thule: What happened?

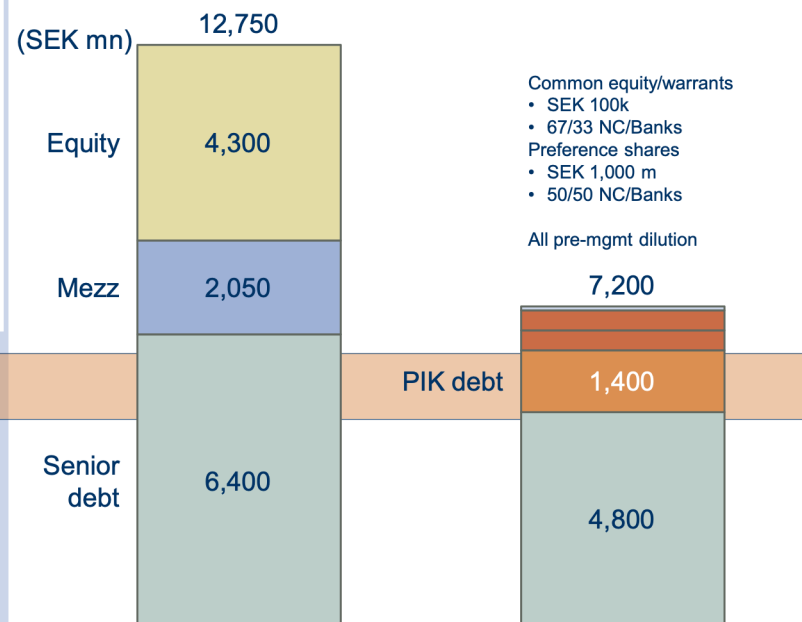
- The capital structure was restructured in Dec 2008 following lengthy negotiations with lenders
 - Original Nordic Capital and GS investment written off
 - Senior banks converted SEK 500 mn of debt to preference shares
 - NC Fund VI invested SEK 500 mn (EUR 47 mn) in preference shares and common equity
 - Part of the senior debt was converted to PIK
 - Remaining debt had less restrictive covenants and amortization
- LP vote agreed increase of single investment limit
 - Thule Group increased from 20.2% of fund VI to 22.7%
- Result:
 - Annual cash debt service decreased by 65% from c. SEK 850 mn to c. SEK 300 mn
 - SEK 1.2bn of liquidity

Thule Group» Financial restructuring

The value of the Thule Group was less than the outstanding senior debt in 2008-H2 on a multiple basis



Value (illustr.)



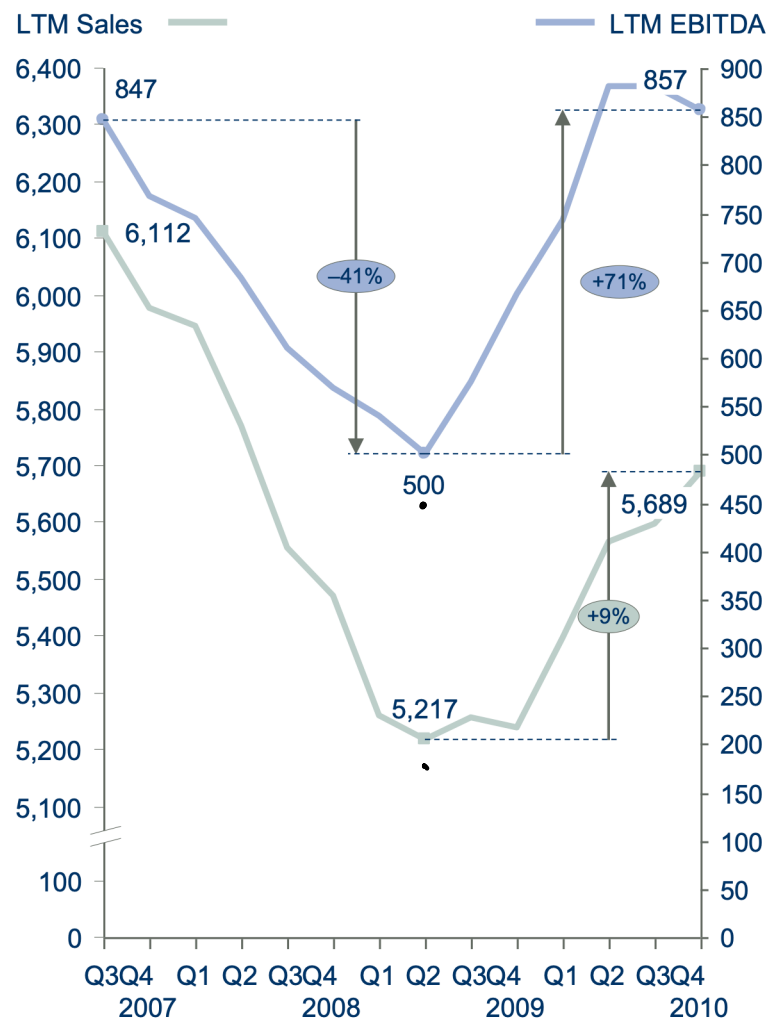
Original investment July 2007 New structure December 2008

Cash-paying loans:	8,450	4,250
Cash interest:	725	~300
Amortisations:	100-150 per year	No amortisations 2009-2011
Revolver:	700	1,125
Headroom covenants:	15%	35-55%
Cash debt service	~850	~300

Result of turnaround was 'V'-shaped

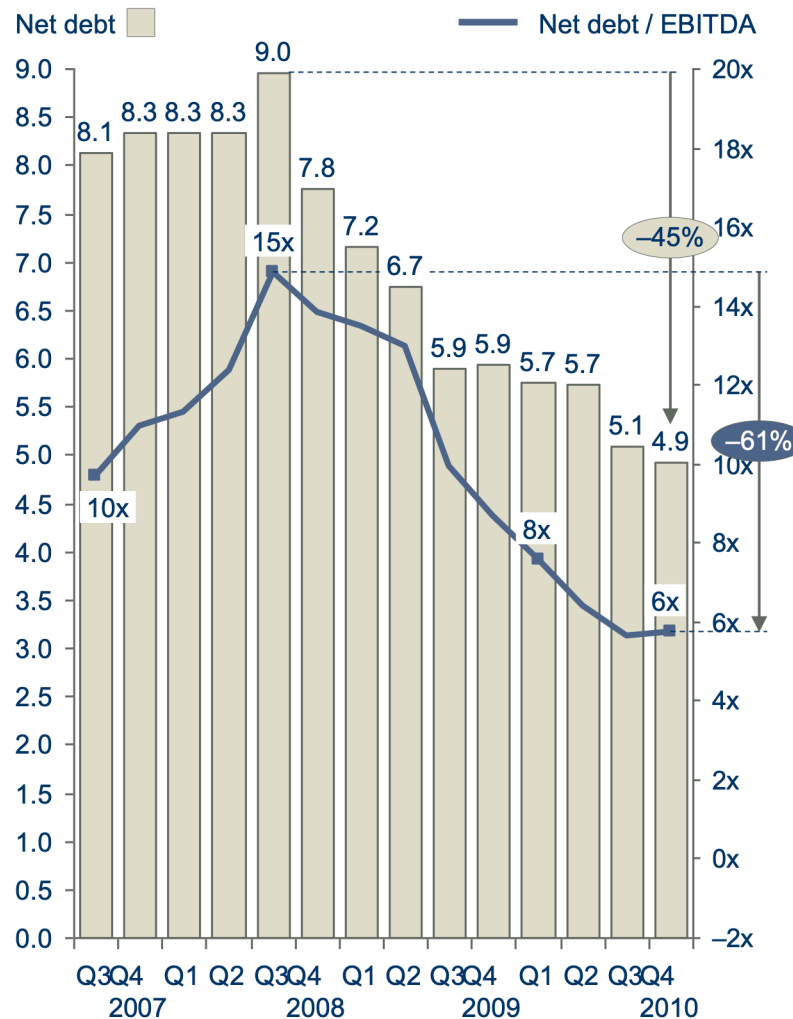
Net sales and adj. EBITDA

SEK mn



Net debt and d:o leverage ratio

SEK bn



- In October 2010, NC Fund VII invested an additional ~SEK 650M in Thule to buy out the bank preferred and common stock (at 7.9x 2010E EBITDA)
- In November 2014, Thule went public on the Stockholm stock exchange
 - Over the the next few years, NC gradually sold their stake in the market
- Overall, NC made a modest, but positive, return on Thule
 - Given that return in fall 2008 was -100%...
 - 2008 and 2010 investments considered to be very profitable to NC

Key Financials¹

For the Fiscal Period Ending	12 months Dec-31-2012A	12 months Dec-31-2013A	12 months Dec-31-2014A	LTM ² 12 months Mar-31-2015A	12 months† Dec-31-2015E
Currency	SEK	SEK	SEK	SEK	SEK
Total Revenue	4,362.0	4,331.0	4,693.0	4,984.0	5,586.12
<i>Growth Over Prior Year</i>	5.8%	(0.7%)	8.4%	NA	19.03%
Gross Profit	1,658.0	1,616.0	1,832.0	1,940.0	-
<i>Margin %</i>	38.0%	37.3%	39.0%	38.9%	-
EBITDA	635.0	644.0	678.0	733.0	921.98
<i>Margin %</i>	14.6%	14.9%	14.4%	14.7%	16.50%
EBIT	542.0	520.0	594.0	646.0	-
<i>Margin %</i>	12.4%	12.0%	12.7%	13.0%	-
Earnings from Cont. Ops.	249.0	299.0	200.0	265.0	-
<i>Margin %</i>	5.7%	6.9%	4.3%	5.3%	-
Net Income	268.0	61.0	(140.0)	(88.0)	-
<i>Margin %</i>	6.1%	1.4%	(3.0%)	(1.8%)	-

Why this case?

- Example of private equity buyouts
- The effect of leverage in LBOs
- Financial restructuring
- Stakeholders in LBOs
- PE operational value-added

Private equity leveraged buyouts

- Acquisitions of (public or private) firms by professional private equity funds
 - Traded companies ("Public-to-Private" deals)
 - Acquire division of another company (e.g. Yell)
 - Independent private (e.g. family owned) company
- Private equity funds
 - Structured as limited partnerships
 - » Institutional investors = LPs
 - » Private equity firm = GP
 - Limited life (10-12 years) after which fund is liquidated
 - » GP compensated by 20% "carry" on profits + 1-3% management fee on fund capital

- 
- For the first 3-5 years, PE fund capital is invested in a number of companies
 - Acquisitions funded partly with equity from fund, partly with debt from banks & bonds
 - Typical financed by 25-50% equity, 50-75% debt

 - For a given acquisition, fund typically has a 3-6 year investment horizon
 - Achieve strategic goals and add value
 - Exit thereafter through IPO or trade-sale (or, if unlucky, write-off/bankruptcy)

How do PE investors increase company value?

- Increase profits (“increase EBITDA”) through engineering (Kaplan and Strömberg, 2009)
 - Governance engineering:
 - » active ownership: small, active board, controlling equity stake, equity incentives to mgmt
 - Financial engineering:
 - » tax shields, financing solutions that lower cost of capital
 - Operating engineering:
 - » help formulate strategies, make acquisitions/divestments, recruit/strengthen management, facilitate expansion, bring in experts on operating issues
 - » Operating engineering has become increasingly important as buyout market has matured and become more competitive
- Buy low/sell high (“increase EV/EBITDA-multiple”):
 - » buy at low valuation (e.g. proprietary dealflow), sell at higher valuation (make co more attractive to investors, make “IPO-able”)
 - » E.g. “buy-and-build”: buy several small companies @ low valuation, merge into large company with high valuation

Examples of PE engineering in Thule

■ Governance:

- Putting industry experts on the board
- Management equity incentive plan
- Transmitting sense of urgency to management and organization

■ Operational:

- Strategic refocusing from auto accessories to branded consumer goods
- Implement better business practices
 - » Lean manufacturing, purchasing, cutting overhead, focusing organization etc
- Improving working capital efficiency
- M&A

■ Financial:

- Managing the financial restructuring

- Do PE owners improve company performance, growth, productivity? Research indicates yes.
- Do investments in PE funds beat the market? Evidence has been more mixed.
 - Recent studies show that PE funds have outperformed S&P500 by ~3% on average.
 - » I.e. Investors get compensated for illiquidity
 - However:
 - » Large differences across fund vintage years. Higher PE fundraising is associated with lower PE returns
- Still, a substantial fraction of PE value added do not go to the investors in the funds
 - Because of PE fund fees
 - » 6-7% p.a. if combine all fees and carry
 - Because of PE funds overpaying for transactions
 - » I.e. selling shareholders get expected value-added.

Reason (1) why hurdle rates $\geq 20\%$

- Since the GP is considering *leveraged equity* cash flows, rather than the CCFs that accrue to all the *assets*, we should use R_E , which is higher than R_A (and R_{WACC}):

$$R_E = R_A + \frac{D}{E} (R_A - R_D)$$

- Picking some “realistic” values, e.g. $R_A = 9\%$, $R_D = 6.5\%$, 70% leverage $\rightarrow D/E = 0.70/0.30 = 2.33$ we get

$$R_E = 9\% + 2.33 \cdot (9\% - 6.5\%) = 15\%$$

Reason (2) why hurdle rates $\geq 20\%$

- Since we are charging 2/20 fees to investors, we need to deliver sufficiently high *net* returns.
 - As you will see in the Excellere case, the difference between gross (before fees) and net (after fees) returns is on the order of 5-7%
- Also, LPs require an additional premium for liquidity
 - Lock in their money for maybe 10 years $\rightarrow$ require compensation for this; otherwise might as well invest in public equity instead.
 - There are no universally accepted models for liquidity premia, but it is not uncommon for LPs to have return targets 1-3% above public equities for illiquid assets.
- E.g. if compensation for fees=6%, compensation for illiquidity = 3%, and $R_E = 12.4\%$ from before:

$$R_E^{\text{adj}} = 15\% + 3\% + 6\% = 24\%$$

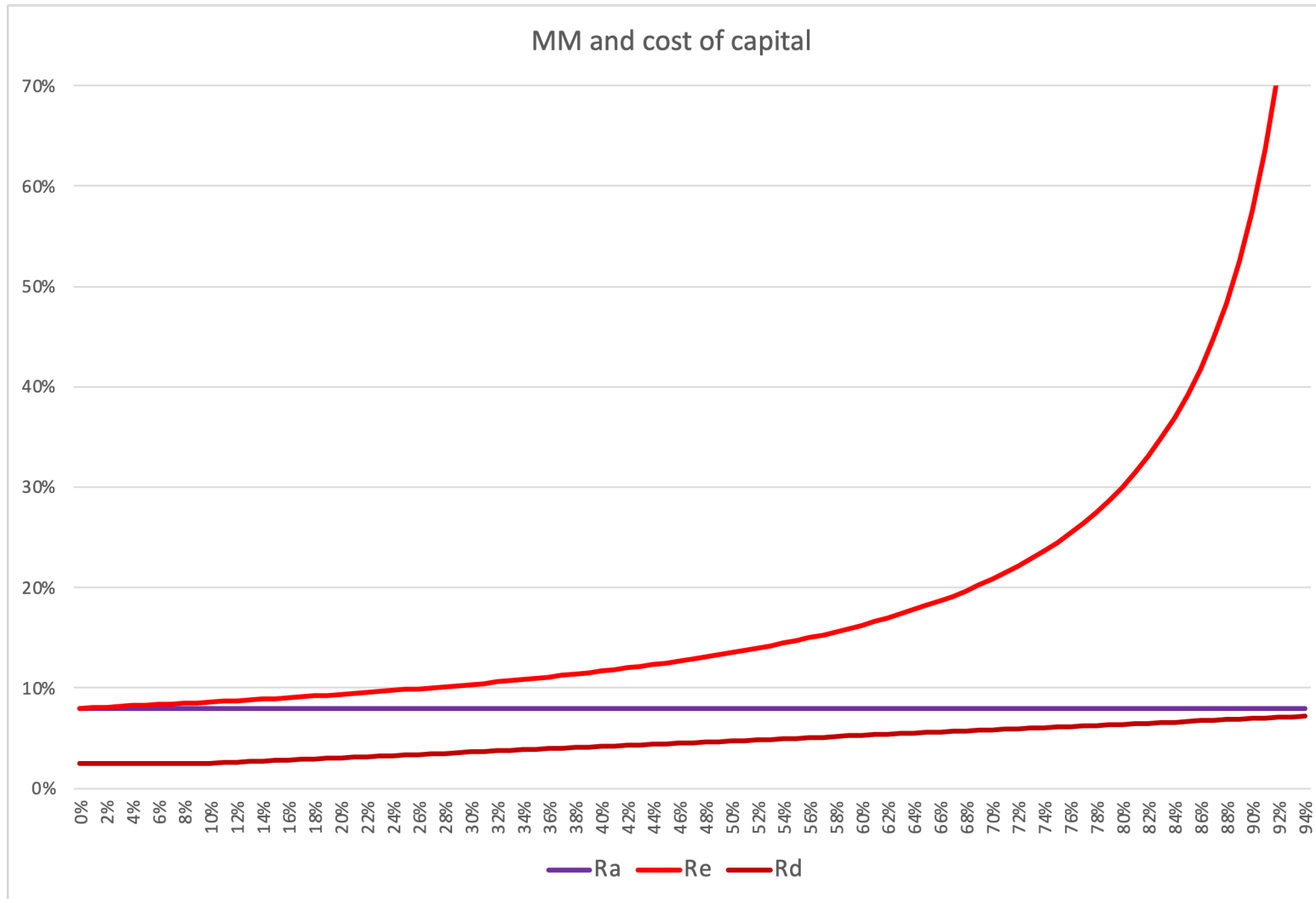
Leverage in LBOs

- Jensen's view: Leveraged buyouts optimize incentive benefits of debt
 - Trade-off against costs of financial distress
- In this case:
 - On the one hand: without leverage, no distress
 - On the other hand: without leverage, no operational change (until too late?)
 - » Would not have brought in Alix, cut costs, etc at such an early stage?

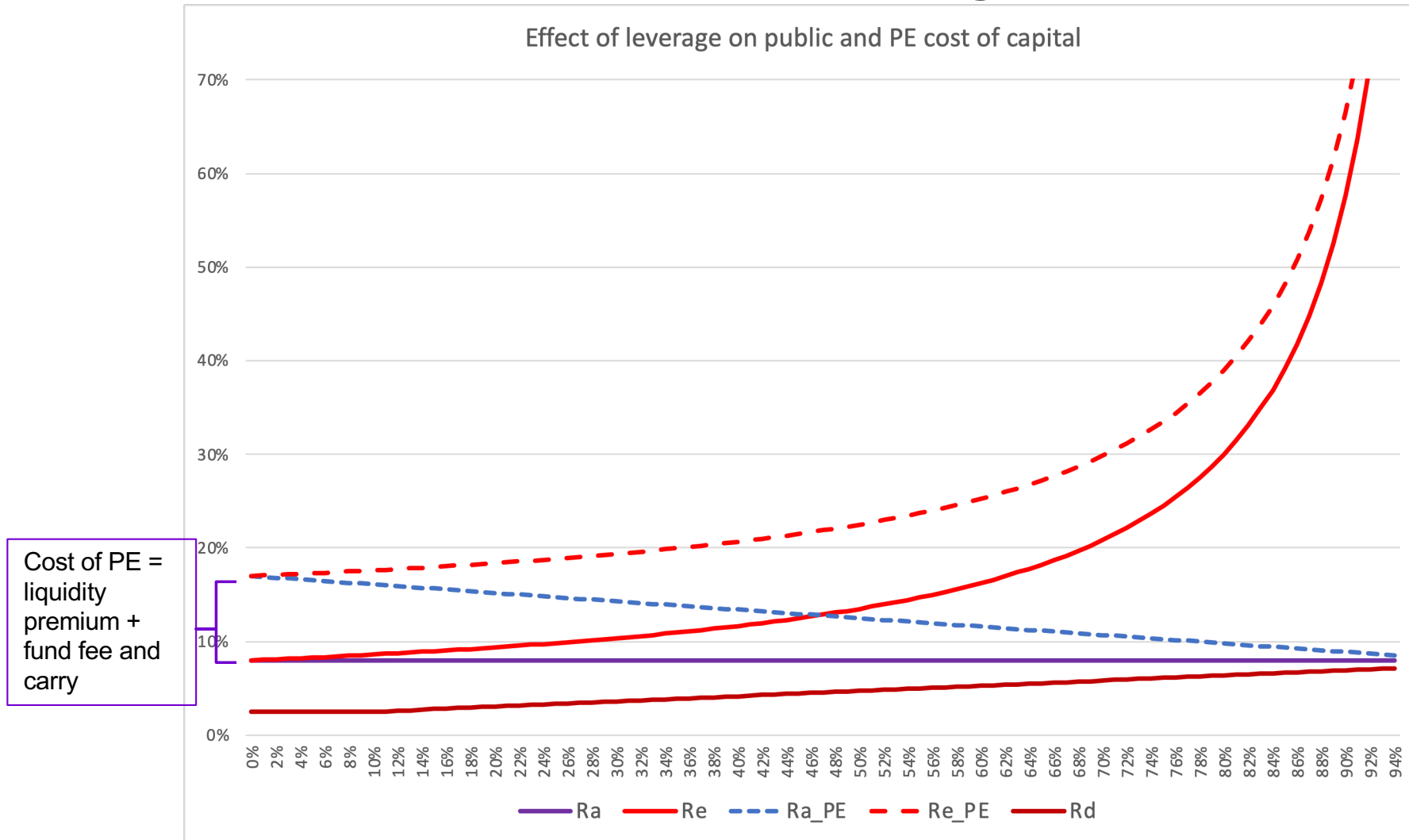
■ But: trade-off model cannot explain LBO leverage

- Rather, PE funds take on as much as leverage as they can get (See Axelson, Jenkinson, Strömberg, Weisbach, 2013)
 - » Leverage vary with market conditions
 - » Higher acquisition prices and lower subsequent returns paid for deals with more leverage
- Why?
 - » PE is more expensive than public equity because
 - LPs will demand a liquidity premium over public equity (~3%)
 - LPs need to be compensated for fee and carry (~6%) to get sufficient net-of-fee returns
 - » Debt financing does not have these additional costs
- Debt is indeed cheaper than private equity
- LBO “optimal” leverage is higher than for listed companies

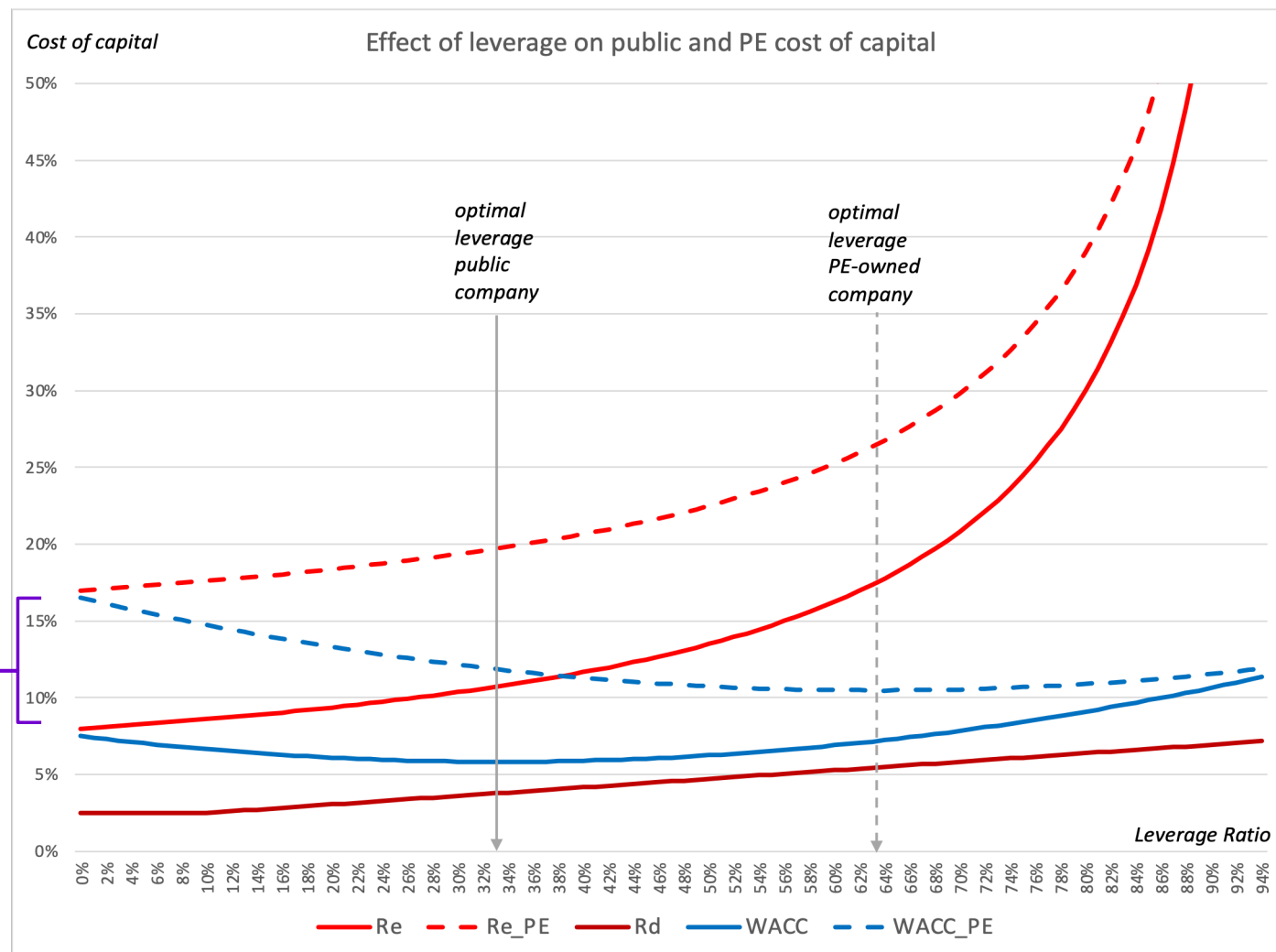
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With costs and benefits of debt ($PVTS + PV(Inc) - PV(COFD)$), optimal leverage level higher in PE



■ Also: COFD lower for PE-backed companies
(Hotchkiss, Smith, Strömberg, 2014)

- Can more easily infuse more equity to avoid distress and/or facilitate restructuring
- Cares about reputation with lenders (and others) → higher incentive to support company

■ Similar to “internal capital market” of conglomerate

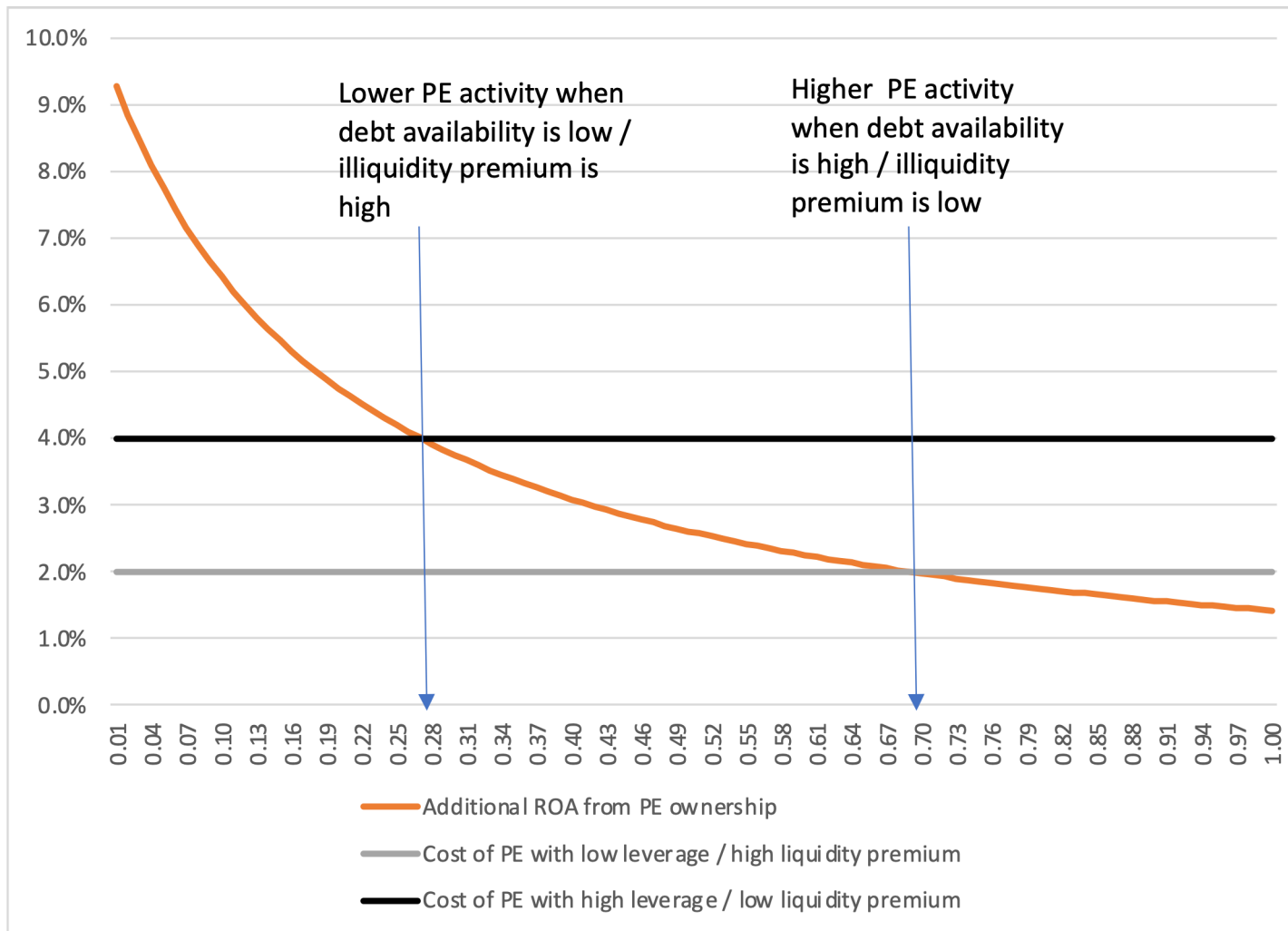
- Can deploy remaining fund equity capital in the portfolio companies that become financially distress
- Major difference from conglomerates: cannot transfer excess cash from one portfolio company to another
 - » → avoid the inefficient cross-subsidization that sometimes happen in conglomerates

LBO leverage and cycles

■ Private equity market cycles

- Investment in private equity increase with high past returns
- Buyout activity increase with low interest rates / hot lending markets
- Most active buyout markets → lower subsequent returns
 - » E.g. 1980's
- After tech crash in 2001: relatively cold market
 - » Great returns for 2002-2003 vintages
- Led to fundraising boom 2006 - 2007.
 - » Returns for these vintages relatively poor

Cycles consistent with changing debt markets and market liquidity premium



“Cold markets”: only firms with highest potential value improvements are viable targets → less activity, lower leverage, higher average productivity gains

“Hot markets”: firms with less value improvements are also viable targets → higher activity, higher leverage, lower average productivity gains

Financial restructuring

- Important to realize difference between:
 - Technical default or breaking covenant
 - Missing debt payment
 - Chapter 11 / Financial restructuring.
 - Liquidation. (Chapter 7)

- I.e.: Financial distress $\neq$ “Game Over”
 - When there is potential gains from saving the firm, there are large incentives to resolve distress through bargaining
 - E.g. debt makes some concessions (e.g. extend maturities, exchange some debt for equity) in exchange for some concessions from equity (e.g. pay higher interest, give up some equity)

- Resolving financial distress is complicated by conflicts of interest among different claimants
 - Equity has incentive is to go for a riskier strategy / get upside
 - Debt, on the other hand, has an incentive to go for a safer strategy
 - » Banks would rather downsize / liquidate than increase revenues

- PE fund can serve mitigating role and facilitate restructuring
 - Long-term reputation with lenders important
 - Can serve as intermediary between bank and management
 - Has more incentive to put in new money than other equityholders – less debt overhang

■ In this case, more complicated

- Mezzanine lender is credible new owner of firm – want to take over equity
- Banks want to avoid taking credit losses during financial crisis

■ Also important to consider other stakeholders

- Management: how stay motivated?
- Suppliers and customers: will company be there in the future? Could become self-fulfilling prophecy
- LP's: concerned about throwing good money after bad?
- Fellow GPs: how can I convince my partners in investment committee to give company one more chance?